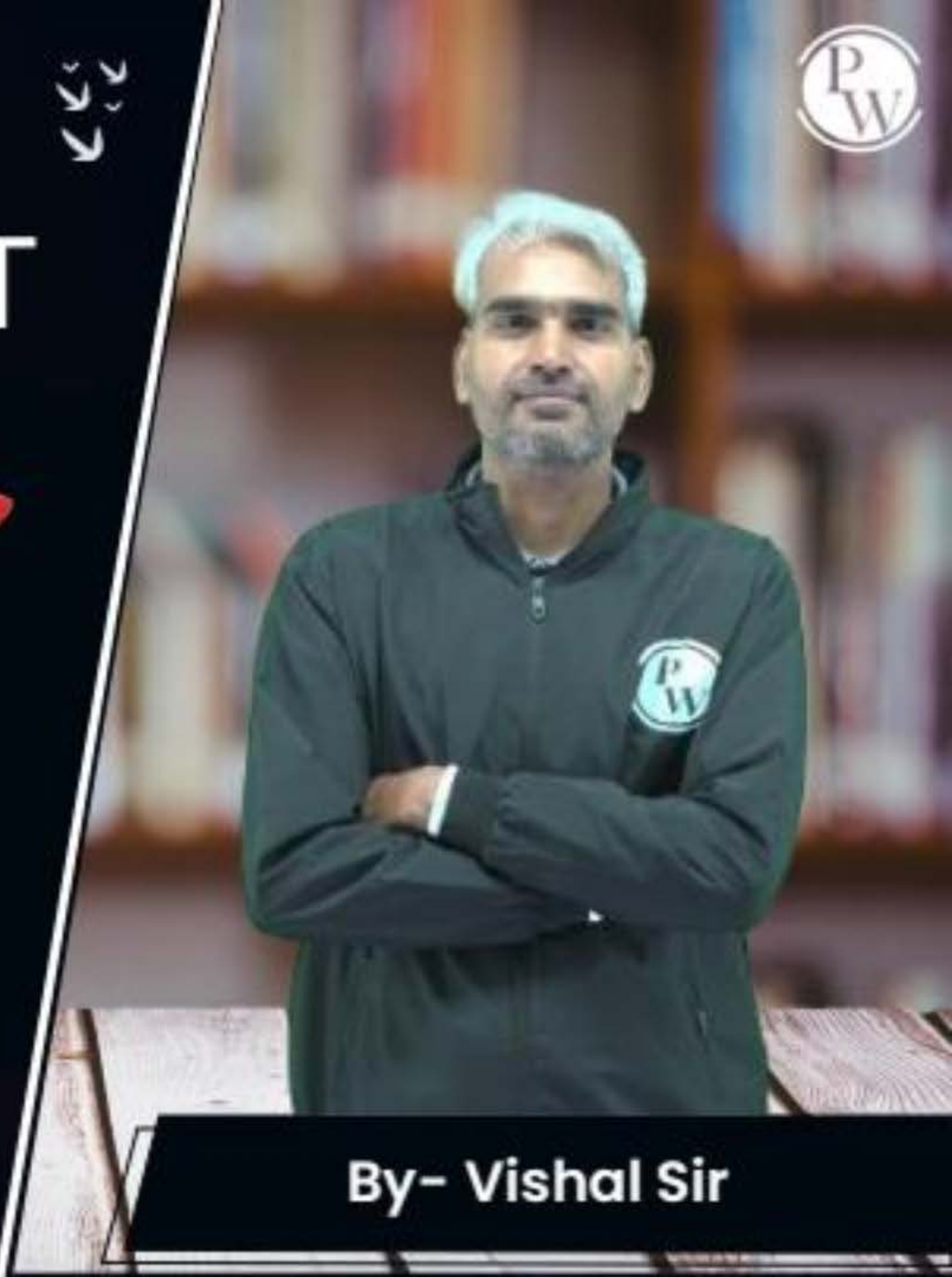
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



- Topic: Super key
- Topic: Closure of an attribute set

Slide

Topics to be Covered



Topic — Identification of candidate keys in a relation

Topic — Membership test

Slide



Topic : Closure of an attribute set

- Closure of an attribute set X (i.e., X^+) can be defined as set of all the attributes that can be functionally determined from attributes of set X.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

A→B

B→C

C→D

$(A)^+ = \{A, B, C, D\}$

$(AB)^+ = \{A, B, C, D\}$

$(AC)^+ = \{A, B, C, D\}$

$(AD)^+ = \{A, B, C, D\}$

$(ABC)^+ = \{A, B, C, D\}$

$(ABD)^+ = \{A, B, C, D\}$

$(ACD)^+ = \{A, B, C, D\}$

$(ABCD)^+ = \{A, B, C, D\}$

$(B)^+ = \{B, C, D\}$

$(BC)^+ = \{B, C, D\}$

$(BD)^+ = \{B, C, D\}$

$(BCD)^+ = \{B, C, D\}$

$(C)^+ = \{C, D\}$

$(CD)^+ = \{C, D\}$

$(D)^+ = \{D\}$

#e.g. Consider the following FD set

$F = \{ AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A\}$

find the closure of following set of attributes.

(i) $\{C,F\}^+ = \{C, F, G, E, A, D\}$

(ii) $\{B,G\}^+ = \{B, G, A, C, D\}$

(iii) $\{A,F\}^+ = \{A, F, D, E\}$

(iv) $\{A,B\}^+ = \{A, B, C, D, G\}$



Topic : Super key

- Let R be the relational schema, and let X be some set of attributes over relation R. If X^+ Contains all attributes of relation R, then X is called super key of relation R.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

A→B

B→C

C→D

$(A)^+ = \{A, B, C, D\}$

$(AB)^+ = \{A, B, C, D\}$

$(AC)^+ = \{A, B, C, D\}$

$(AD)^+ = \{A, B, C, D\}$

$(ABC)^+ = \{A, B, C, D\}$

$(ABD)^+ = \{A, B, C, D\}$

$(ACD)^+ = \{A, B, C, D\}$

$(ABCD)^+ = \{A, B, C, D\}$

Closure Contains all attributes of the relation.

∴ All eight sets of attributes are Super keys of relation R

$(B)^+ = \{B, C, D\}$

$(BC)^+ = \{B, C, D\}$

$(BD)^+ = \{B, C, D\}$

$(BCD)^+ = \{B, C, D\}$

$(C)^+ = \{C, D\}$

$(CD)^+ = \{C, D\}$

$(D)^+ = \{D\}$

Proper subset :- For a given set A, any subset of set A, except set A itself is called Proper subset of set A.

eg: let $A = \{a, b, c\}$

Subsets of set A are : $\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}$

$\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}$ — Proper subsets of $\{a,b,c\}$

$\{a,b,c\}$ — it is equal i.e. Not a Proper subset



Topic : Candidate key (Minimal Super key)

- Let R be the relational schema, and let X be the super key of relation R.
- If no proper subset of X is a super key the X is minimal super key i.e., X is Candidate key

eg: let $(A,B)^+$ = {All attributes of Relⁿ R}
∴ AB is a Super key of R

Non-Empty Proper subsets of {A,B} are

{A}, let $(A)^+$ = Not all attributes of R
∴ A is not a S.K.

{B}, Let $(B)^+$ = Not all attributes of R
∴ B is not a S.K

No Proper subset of {A,B} is a Superkey
∴ (AB) is minimal S.K
Hence, (AB) is C.K

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$AB \rightarrow C$,

$B \rightarrow E$,

$C \rightarrow D$

Find the Candidate key of R.

Note:- Attributes that are not present in R.H.S. of any of the FD of FD set are Called "Essential attributes"

"Every Essential attribute must be present in Every Candidate key"

A & B are essential attributes,

$(AB)^+ = \{A, B, C, D, E\}$ — all attributes

∴ "AB is a Superkey"

Both A & B are essential ∴ No one can be removed from the key

Hence, AB is a C.K

Non-empty proper subsets of AB

$\{A\}^+ = \{A\}$ ∴ A is not a S.K

$\{B\}^+ = \{B, E\}$ ∴ B is not a S.K

No proper subset of AB is a S.K

Hence AB is a C.K

In the above example **AB** is a C.K.

∴ Prime Attributes = {A, B}

* If there is no FD in the FD set of the relation, such that any of the prime attribute appears in the R.H.S. part of the FD, then that relation will have only one Candidate Key.

A & B does not appear in R.H.S. part of any FD

∴ Relation will have only One Candidate Key

and Non Prime Attributes = {C, D, E}

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$AB \rightarrow C$,

$B \rightarrow E$,

$C \rightarrow D$

$E \rightarrow A$

Find the Candidate key of R.

$(AB)^+ = \{A, B, C, D, E\}$ all attributes ∴ AB is a Super key

Non-Empty Proper subsets of {A, B}

$\{A\}^+ = \{A\}$

A is not a S.K

$\{B\}^+ = \{B, E, A, C, D\}$ all attributes ∴ B is a S.K

At least one proper subset of AB is a S.K.

∴ AB is not minimal. Hence AB is just a Super key not a C.K.

B is a new Super key

↳ A key with a single attribute is always a C.K.

∴ B is a Candidate key

B is essential attribute

$(B)^+ = \{B, E, A, C, D\}$ all attribute ∴ B is S.K

B is a S.K with single attribute

∴ B is a C.K.

* In the above example **B** is a C.K

Prime attribute = {B}

Prime attribute B is not present in the R.H.S. part of any FD, ∴ There will be only one C.K in the relation

Hence, Non-prime Attribute = {A, C, D, E}



Topic : Note

* If there exists any FD in the relation of type,

$X \longrightarrow$ Any of the prime Attribute

then we can replace this prime attribute in that Candidate key by "X", and we will find a new Super key after replacement

* If there exists any FD in the FD set such that any of the prime attribute appears in the R.H.S part of any of the FD, then relation will have more than one Candidate key

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

$AB \rightarrow CD$, $\equiv$ $AB \rightarrow C$, $AB \rightarrow D$ (Splitting Rule)

$D \rightarrow A$

Find all the Candidate keys of R.

$(AB)^+ = \{A, B, C, D\}$ — all attributes

$\therefore$ AB is S.K → Non Empty Proper subset of $\{A, B\}$

$\{A\}^+ = \{A\}$ Not a S.K

$\{B\}^+ = \{B\}$ Not a S.K

No proper subset of $\{A, B\}$ is a S.K

$\therefore$ **AB** is a C.K

Prime attributes = $\{A, B\}$

in $D \rightarrow A$

R.H.S. is prime attribute

$\therefore$ We can replace A by D to get a new S.K.

$AB \rightarrow D$

$\therefore$ Replace D by AB

DB is a S.K

$(D)^+ = \{D, A\}$

$(B)^+ = \{B\}$

No proper subset of DB is a Super Key

$\therefore$ **DB** is also a C.K

Prime Attribute = $\{A, B, D\}$

Essential attribute is B

$(B)^+ = \{B\}$

$\therefore$ Not a Super Key

$\therefore$ B must be present in Every C.K

- In the above example (AB), and (DB) are the only two Candidate Key.

∴ Prime Attribute = {A, B, D}

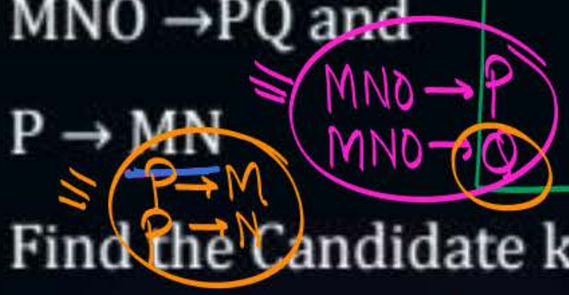
Non-prime attribute = {C}

#Q. Assume a relation R (M, N, O, P, Q) that has the following functional dependencies:

MNO → PQ and

P → MN

Find the Candidate keys of R.



$(MNO)^+ = \{M, N, O, P, Q\}$ — all attributes ∴ MNO is a S.K

Non-Empty proper subsets

$(MO)^+ = \{M, O\}$

$(NO)^+ = \{N, O\}$

No proper subset of "MNO" is a S.K

∴ MNO is a C.K

{Don't need to check for other proper subsets because 'O' must be present}

O is prime attribute

$(O)^+ = \{O\}$

MNO is a C.K

Prime Attribute = {M, N, O}

MNO → P

∴ Replace "P" by "MNO"

P → MN — both are prime attributes of same C.K

∴ (MN) Can be replaced by P

PO is a S.K as well as is a C.K {'O' is Essential}

Prime attribute = {M, N, O, P}

(MNO) & (PO) are the only two Candidate Keys

Prime Attribute = {M, N, O, P}

Non-prime Attribute = {Q}

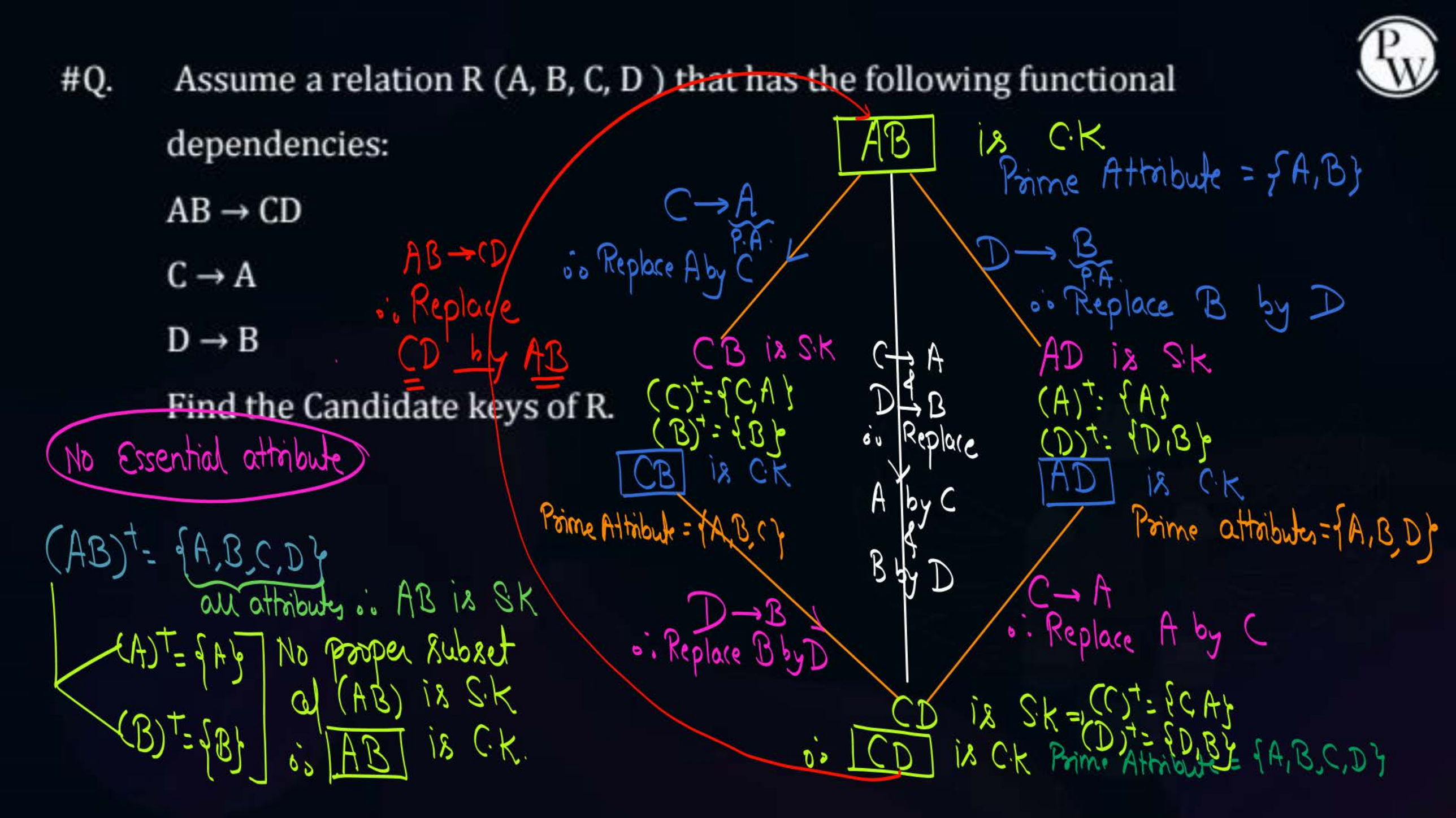
#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

$AB \rightarrow CD$

$C \rightarrow A$

$D \rightarrow B$

Find the Candidate keys of R.



No Essential attribute

$(AB)^+ = \{A, B, C, D\}$ all attributes ∴ AB is S.K

$(A)^+ = \{A\}$

$(B)^+ = \{B\}$

No proper subset of (AB) is S.K

∴ AB is C.K.

In the above example, (AB), (CD), (CB) & (AD) are four candidate keys of the relation

Prime Attributes = {A, B, C, D}

Non-prime Attributes = { } (Empty), i.e, No non-prime attribute

#Q. Assume a relation R (A, B, C, D, E, H) that has the following functional dependencies:

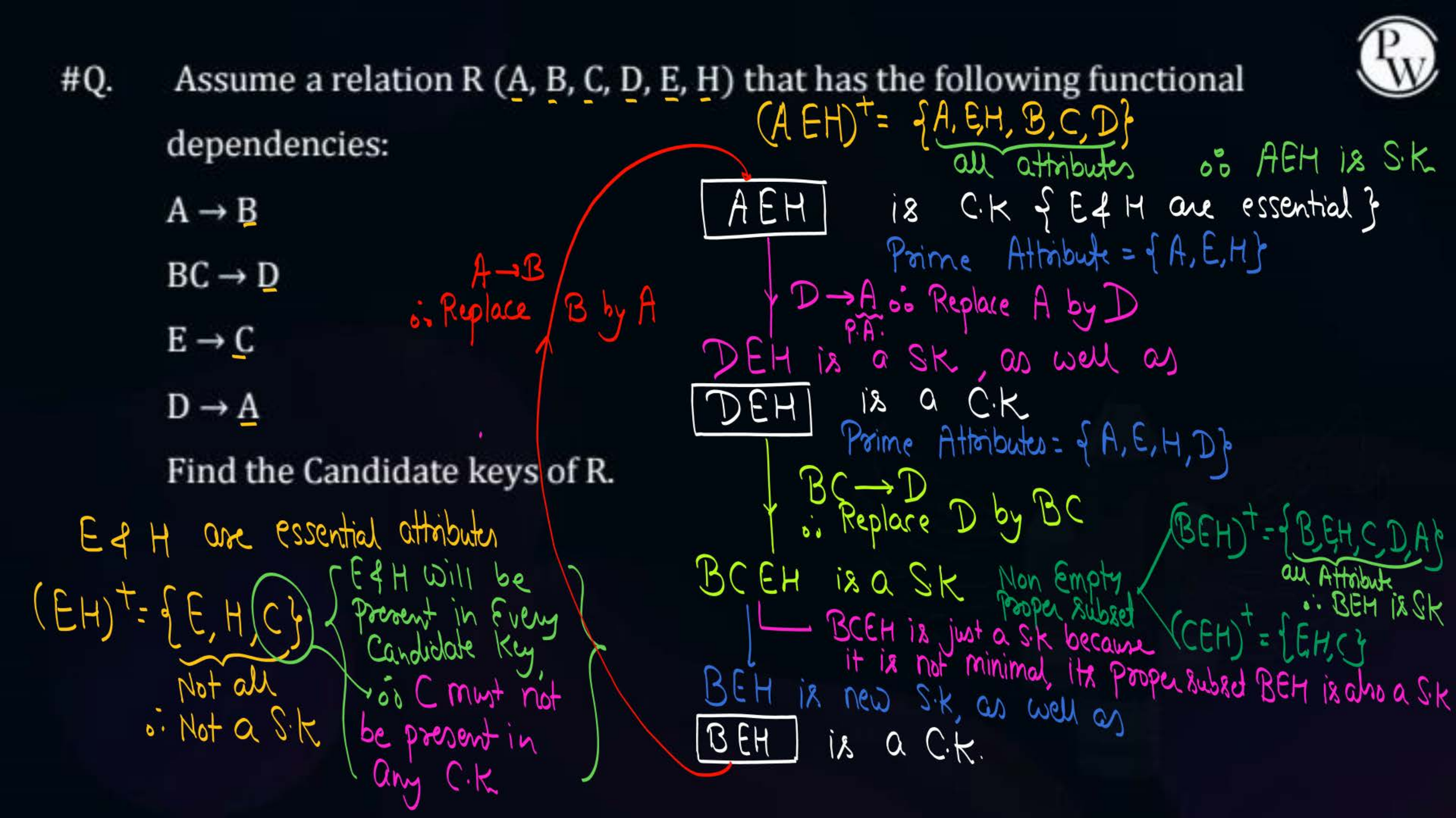
$A \rightarrow B$

$BC \rightarrow D$

$E \rightarrow C$

$D \rightarrow A$

Find the Candidate keys of R.



In the above example,
(AEH), (DEH) & (BEH) are three Candidate keys

Prime Attribute = {A, B, D, E, H}

Non-prime attribute = {C}



2 mins Summary

Topic Identification of candidate keys in a relation

Topic Membership test

THANK - YOU